

Executive Summary: Regression Analysis

TikTok claims classification project

OVERVIEW

The TikTok data team is working on a machine learning model designed to classify claims in user submissions. Analysis has revealed that verified users are significantly more likely to share opinions. Given the importance of distinguishing between claims and opinions, understanding the posting behavior of verified accounts is essential. To achieve this, a logistic regression model has been developed to predict `verified_status`, enabling the identification of patterns in user behavior.

PROJECT STATUS

The TikTok data team is developing a machine learning model to classify claims in user submissions. Verified users are more likely to share opinions, making it crucial to understand their behavior. A logistic regression model was chosen to predict `verified_status`, given its relevance to video content and suitability for classification.

The model achieved 74% precision for "verified" and 61% for "not verified," with recall of 46% and 84%, respectively. Its overall accuracy was 65%, highlighting its strength in identifying "not verified" users while struggling with recall for "verified" ones. The F1 score was 66%, offering key classification insights.

NEXT STEPS

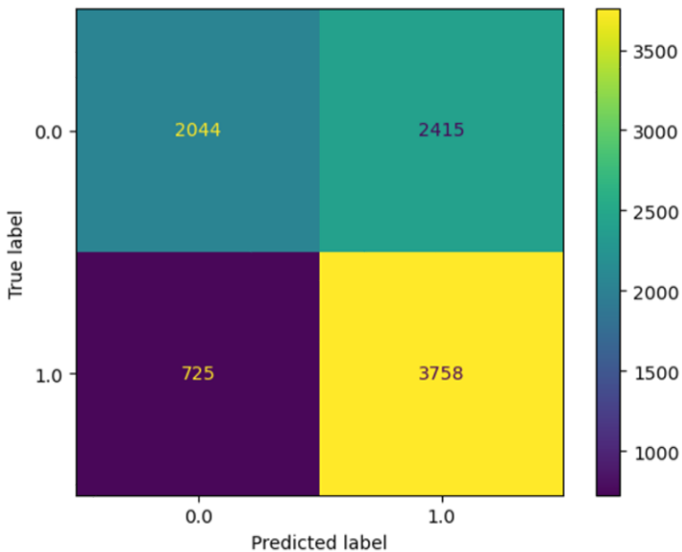
The next step is to build a classification model to predict the status of user claims, which is the final project and original expectation from the TikTok team. With the available data, we now have sufficient context on user behavior to effectively analyze the model's results.

KEY INSIGHTS

The logistic regression model estimates that longer videos are associated with higher odds of the user being verified, with coefficient of $8.607893e-03$.

Other video features have small coefficients, indicating their weak association with verified status. Therefore, apart from video length, other features do not appear to be strongly linked to verified status.

Confusion matrix for logistic regression model



0.0 = number of videos posted by unverified accounts.
1.0 = number of videos posted by verified accounts.